WOOD, FIELD, RAILINGS AND RUBRAILS - SYSTEM WFR: Alkyd System Wood Railings and Rubrails

MoT Usage: This system is designed for use in a field over coating situation for wooden railings and rubrails, where the original alkyd system is intact or where loose paint can be removed by hand/power tools.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Can/CGSB 1-GP-189M	None	If Used, Midcoat Material	**White/ Grey** CAN/CGSB 1.59M 89	None	White latex CAN/CGSB 1.138-93 White alkyd CAN/CGSB 1.59M89 For guardrails: Yellow/White alkyd CAN/CGSB 1.28M89 For sign/posts: Green alkyd Yellow alkyd White alkyd CAN/CGSB 1.59M89
SSPC- SP 2 "Hand Tool Cleaning" or SP 3 "Power Tool Cleaning"					

Notes for use: None

WOOD, FIELD TOUCHUP - SYSTEM WFT: Alkyd System for Touchup on Wood Structures

MoT Usage: This system is designed for use in a field over coating situation where the original alkyd system is intact or where loose rust/paint can be removed by hand/power tools.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
DOH A-3 CGSB 1-GP-18 9M	None	If Used Midcoat Material	White/ Grey CAN/CGSB 1.5 9M89	None	White latex CAN/CGSB 1.138-93 White alkyd CAN/CGSB 1.59M89 **For guardrails:** Yellow/White alkyd CAN/CGSB 1.28M89 **For sign/posts:** Green alkyd Yellow alkyd White alkyd CAN/CGSB 1.59M89
SSPC- SP 2 "Hand Tool Cleaning" or SP 3 "Power Tool Cleaning"					

Notes for use: None

SECTION 312

STEEL TRAFFIC BARRIERS AND WOOD POSTS

312.01 Scope - This Section covers the quality and manufacture of steel beam guardrails, wood posts and accessory material for use as traffic barriers.

The layout and installation of materials specified in this Section are generally indicated on the SS Drawings of the SP312 series.

DBSS 604 covers the installation of the material of this Section together with the construction of ancillary work

312.02 Standards of Reference - Standards of various authorities are referred to herein generally in sufficient detail for the purpose of this Section. For detailed testing and other requirements reference to the Standards may be necessary from actual copies obtainable as follows:

CSA	Canadian Standards Association: Standards Sales, 178 Rexdale Blvd, Rexdale, ON M9W 1R3
ASTM:	American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103
ARTBA:	American Road and Transportation Builders Association, 525 School Street SW, Washington DC 20024
AASHTO:	American Association of State Highway and Transportation Officials, 444 North Capitol Street N.W., Suite 225, Washington, DC 20001.

312.03 General - Steel traffic barriers shall consist of rail sections fabricated for installation to develop a continuous beam strength with the necessary safety end feature components.

All rail sections and other components shall match the design profiles and dimensions of the AASHTO/ARTBA requirements cited below for full interchangeability of similar components regardless of the source of manufacture.

The name and/or trademark of the manufacturer and the metal thickness shall be clearly and permanently marked on each component clear of the splicing overlap and on the opposite face to the traffic side.

312.04 Steel Beam Guardrail - Guardrail shall be manufactured from open hearth, electrical furnace or basic oxygen semi-spring steel sheet galvanized before forming (or afterwards as may be specified or ordered for authorized special installations) all in general accordance with the AASHTO Standard Designation M-180 "Corrugated Sheet Steel Beams for Highway Guardrail" and shall conform to the following detail requirements.

Note: Design and part numbers are taken from the ARTBA Technical Bulletin #268-B, current edition (unless stated to be BCH Part numbers) and are reproduced on SS Drawings 6-SP312 through 10-SP312.

312.04.01 Beam Classification (According to the AASHTO Standard) - The shape, class and type together with the ARTBA design and part numbers shall be as specified in the Contract and the Design, and in accordance with the SS Drawing SP312 series.

i) Shape - W-Beam are for ARTBA Design G4 and MB4 installations and Thrie-Beam for Design G9 and MB9 installation and both shall be accurately formed to profile, dimensions and tolerances of the AASHTO Standard with overall cross sections of 311 mm x 85 mm and 508 mm x 85 mm respectively (ARTA Designation RE-3 and RE-63).

ii) Class and Type - Unless otherwise specified or ordered, beams shall be Class A, Type 1. Authorized special installations for corrosive conditions and environment of industrial and marine areas may necessitate a Class A, Type 2 requirement.

Note: Type 4 shop primed beams will not normally be specified.

iii) Length - Normally 3.81 m (12' 6") nominal length beams (4.19 m, +0, -75 mm overall) will be required.

Note: 1.9 m (6' 3") and other lengths if specified or ordered may be considered a special order.

iv) Punchings - In addition to joining bolt holes, punchings may be designed as 7.82 m, 3.81 m or 1.9 m (25' 0", 12' 6", 6' 3" respectively), see SS Drawings 1-SP312 and 2-SP312 for requirements.

Note: Punchings at 0.95 m, 1.27 m (3' 1½" and 4' 2") and other centres if specified or ordered may be considered a special order.

v) Curved Rails - For curved rail data see SS Drawing SP312-11.

312.04.02 Metal

i) Properties - Properties of the base metal for beams and any transition section shall conform to the following requirements:

- Yield point minimum: 345 MPa
- Tensile strength minimum: 483 MPa

- Elongation minimum in 50 mm: 12%

Note: End and buffer sections shall have minimum yield point of 227 MPa and minimum tensile strength of 310 MPa.

ii) Sheet Thickness - Class A base metal thickness shall be 2.8 mm nominal (2.67 mm minimum) and for special authorized installations Class B 3.5 mm nominal (3.43 mm minimum).

iii) Sheet Widths - Sheet Widths shall be minimum 483 mm for W-Beams and 749 mm for Thrie-Beams both with a permissible tolerance of minus 3.2 mm.

312.04.03 Zinc Coating - Zinc for coating beams and ancillary sections shall be at least equal to "Prime Western" conforming to CSA Standard HZ2 or AASHTO Standard M120 (ASTM B 6) "Slab Zinc".

The coating shall be firmly bonded to the base metal whether fabrication is from galvanized sheet or galvanizing is specified after fabrication.

The average mass of zinc coating Type 1 (AASHTO M180) shall be 610 g/m² for three specimens but minimum 550 g/m² for any individual specimen being the total amount on both sides of the sheet or beam. When Type 2 is specified or ordered (see DBSS 312.04.01 (ii)) the coating shall have average 1220 g/m² and 1100 g/m² individual minimums.

312.04.04 Quality of Work - Beams shall be punched for splice and post bolts in strict conformity with the AASHTO Standard to the designated number of and centre to centre spacing of posts.

Upon delivery the beams shall be ready for assembly and installation.

Field cutting shall be avoided where practicable. Where necessary, field cutting shall be performed with a cut-off or hack saw; field hole cutting with a drill and/or reamer. Flame or plasma cutting or welding shall not be permitted. All field cuts or holes shall be deburred and treated with two coats of cold galvanizing or zinc-rich coating to CGSB Standard 1-GP-181M.

Warped or otherwise deformed beams will be rejected as will those with injurious defects or excessive roughness of the zinc coating.

Beams for concave and/or convex curved installations with a radius of 45 m or less shall be accurately and smoothly bent in the shop with minimum deformity. The radius of curvature shall be clearly stencilled on the back of each curved rail section.

312.04.05 Inspection - No material shall be shipped prior to inspection and certification by quality management staff.

All materials may be inspected before shipment to the construction site, or at the construction site, or both. The supplier or manufacturer shall provide the necessary facilities to enable the Quality Manager or Ministry Representative to expeditiously examine selected pieces or take specimen samples for testing from the material lot(s) clearly identified for intended supply to the project. Identification shall be such that after inspection and testing the lot may be subsequently identified for acceptance or rejection.

Inspection of material before shipment shall not exclude its subsequent rejection at the construction site if found to fail any requirements of this specification.

312.04.06 Examination - Examination may include the following:

i) Dimension measurements for size and thickness all of which shall be within the tolerance measurements specified by DBSS 312.04.01 (i) and Table 312-A.

TABLE 312-A

GALVANIZED BEAM THICKNESS			
FINISH			
Base Metal	Type 1	Type 2	Tolerance
Class A (2.8 mm nominal)	2.74 mm	2.82 mm	minus 0.23 mm
Class B (3.5 mm nominal)	3.51 mm	3.58 mm	minus 0.25 mm
Note: Tolerance given is for under specified thickness; there is no limit for over thickness.			

ii) Coating generally shall be smooth, free of beading or sharp projections at edges. Coating adherence shall prevent the peeling of any portion of the zinc coating so as to expose the base metal by cutting or prying with a stout knife under considerable pressure.

312.04.07 Testing - Testing shall be carried out according to ASTM A 90 - "Weight of Coating in Zinc Coated . . . . . Articles" (also AASHTO T65) to determine the mass of galvanizing in conformity with the requirements of DBSS 312.04.03. Alternatively magnetic gauge determination in accordance with ASTM E 376 may be permitted.

The testing of the mechanical properties of the base metal shall be confirmed by or carried out by the Quality Manager for compliance with the requirements of DBSS 312.04.02.

312.04.08 Rejection and Retests - Should the beam tested fail to meet the testing requirements specified above, two additional beams as selected at random by the Quality Manager or Ministry Representative shall both meet the test requirements in every respect, otherwise the complete lot represented by the samples will be subject to rejection.

In the event of complete lot, the intended replacement lot shall be satisfactorily tested by a mutually acceptable independent testing agency at the supplier's expense. When the time delay for retesting is unacceptable to the Contractor the order or part thereof will be subject to cancellation.

Rejected material shall be replaced at the Supplier's or Contractor's expense, including shipping charges and removal of rejected material at the construction site, if applicable.

312.05 Guardrail Accessories

312.05.01 Transition Section - Transition section for W-Beam to Thrie-Beam shall be of matching metal thickness and finish and fabricated to the ARTBA design profile (RE-69) to provide a smooth and uniform transition of beam widths and corrugations.

312.05.02 End Sections - End sections shall be of the type specified or ordered as indicated on standard SS Drawing 2-SP312. Double face buffer (ARTBA Designation RE-7) sections for breakaway cable terminal (BCT) assemblies shall be of the stipulated radius and the metal thickness and finish to match the beams. Terminal connectors (RE-8), as return sections of the BCT buffer ends and as normal beam connections (RE-8 or RE-67) to bridge parapets, abutments and the like, the metal thickness shall be Class B with matching finish.

312.05.03 Back-up Plates - Back-up Plates required where steel beams between splices are to be installed to any square or channel section steel posts (standard 305 mm lengths) shall be fabricated of the stipulated lengths from the standard beam section (as or similar to RE-4 or RE-64 ARTBA Designation).

312.05.04 Bolts, Nuts and Washers - Button-head bolts for joining beam sections in the length or to end sections and for fixing beams to posts (ARTBA Designation F-3) and hexagonal-head bolts elsewhere shall be of the stipulated diameter, thread length and/or bolt length.

Bolts generally shall conform to the requirements of ASTM A 307, nuts to ASTM A 563 Grade A and plain washers to ANSI B27.2 Type A requirements, and unless otherwise specified, all galvanized according to ASTM A 153.

Plate washers where required (as ARTBA Designations F-12, F-36 and F-38) shall be of the size and thickness stipulated to ASTM A 36 and galvanized to ASTM A 153 requirements.

312.05.05 Post Sleeve - Post Sleeve for BCT breakaway cable fixing through the wood post shall be a 150 mm length of 60 mm outside diameter galvanized pipe conforming to ASTM A 120 for "standard weight" pipe (ARTBA F-34).

312.05. 06 Cable Assembly - Cable Assembly for BCT installation shall consist of a swaged fitting, 25 mm diameter threaded stud, locking pin, nut and washer at each end of a cable of 19 mm nominal diameter, 6 x 19 wire strand core independent wire rope core, galvanized, right regular lay wire rope with a minimum breaking strength of 200 000 N (45,000 #) and conforming to AASHTO M-30 with the overall assembly length of 1980 mm developing a similar breaking strength (ARTBA F-37, F-35 and complete with plate washers, F-36 one end and F-38 the other).

312.05.07 Anchor Plate - Anchor Plate for the breakaway cable attachment to the steel beam for BCT assemblies shall be fabricated to the ARTBA design profile (RE-71) from steel plate conforming to ASTM A 36 and galvanized according to ASTM A 123 after forming and drilling for and including eight 16 mm diameter hexagonal-head bolts 38 mm long each with nut and 3.5 mm thick plain washer 44 mm outside diameter (ARTBA F-8 and F-13) complying with DBSS 312.05.04.

312.05.08 Anchor Inserts - Anchor Inserts for fixing terminal connector end sections to existing concrete shall be heavy duty hexagonal bolt type expansion anchor "Hilti" size HSL M16/25, or other pre-approved alternative with hardened plain washer minimum 4 mm thick and 57 mm outside diameter with all metal surfaces galvanized.

Anchors of alternative manufacture may be approved by the Ministry Representative provided equivalence of design and effect can be demonstrated.

312.05.09 Steel posts - Where required for installation on concrete, steel posts shall be fabricated from standard structural steel H-beam or bent steel plate to channel or "C" shape and with similar section short length offset "blocks" and heavy base plates and all shall conform to CSA Standard G40.21M type 300 W or ASTM A 570, Grade 45 and/or ASTM A 36.

All welding shall conform to the requirements of the American Welding Society Code AWS D1.1. All welding shall be carried out by qualified operators of a contractor approved by the Canadian Welding Bureau or the American Welding Society to the requirements of CSA Standard W47.1 Division 3 or the AWS Code.

Galvanizing, according to CSA Standard G164 or ASTM A 123, shall be carried out after all punching, drilling, cutting and welding.

Post type, size and length similar to ARTBA Designation P-42 or P-53, complete with connecting bolts (F-8), expansion anchors similar to DBSS 312.05.08 or through bolts and backing plates or other fixing devices shall be as stipulated and/or detailed.

Alternative shape, type and size of post members and fixing devices to those detailed or ordered may be permitted by the Ministry Representative provided equivalence of design and effect can be demonstrated.

312.06 Wood Posts

312.06.01 Square Posts - Square posts and offset blocks shall be Douglas Fir/Hemlock "No. 1, Structural Posts and Timber", graded in conformity with the requirements of NGLA "Standard Grading Rules for Canadian Lumber".

Wanes on any face shall not exceed the following width:

- Above grade (including blocks): 25 mm
- Below grade: 60 mm

being the minimum permissible post width less the portion entirely free of wane.

Posts and blocks shall be supplied in the exact lengths ordered or specified and unless otherwise required on the Design the scantling shall be nominal 200 mm x 200 mm, and pair of 200 mm x 150 mm for each BCT installation, with dressed on four sides minimum dimension of 189 mm x 189 mm and 189 mm x 138 mm respectively and a tolerance of plus 3 mm.

312.06.02 Preservation Treatment - All posts and blocks shall be pressure treated in accordance with CSA Standard 080-M "Wood Preservation".

Unless otherwise specified or stipulated by the Design, the preservatives and retention of preservatives shall conform to the recommendation of CSA Standard 080.14-M Table I "Minimum Retention of Preservatives in Pressure Treated Wood for Highway Construction" under the heading of "Posts-Guardrail..." for square posts and blocks.

Prior to treatment all material shall be air seasoned in accordance with AWPA Standard M1 Paragraph 1.31 to a moisture content of 15% to 22% before pressure treatment. In exceptional circumstances the Designer, Quality Manager and Ministry Representative may permit conditioning by steaming for a total of not more than 6 hours at a temperature not in excess of 115°C.

312.06.03 Inspection - All processing of the material shall be open for inspection by the Quality Manager or the Ministry Representative with free entry to the treating plant while the work is being performed.

No material shall be shipped prior to inspection by quality management staff.

Material inspected before shipment shall not bar its subsequent rejection after delivery if found to fail any requirements of this specification. Rejected material shall be replaced at the Supplier's expense including shipping charges and removal of rejected materials if applicable.

Note: Inspection of material already in bundles ready for shipment shall be considered as "incomplete", and the material will be subject to final inspection only when the bundles are opened immediately prior to use.

SECTION 316

STEEL AND WIRE FENCE MATERIALS

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

316.01 Scope - This Section covers the quality and manufacture of wire fencing material, steel posts, gates and hardware, and various accessory material.

Treated wood fence posts are specified by DBSS 909.

For convenience, chain link fabric necessary for protection on rock slopes is included in this Section.

The uses of materials specified in this Section are generally indicated on Drawings of the SS SP741 series for construction of fences and DBSS 207 for rock protection.

The types of Standard Wire fencing and Chain Link fabric for fencing and protection on rock slopes covered by this Section are designated as follows:

- Type A - Special Wire Fabric Fence for use only on railway rights-of-way
- Type B - Standard Wire Fabric Fence
- Type C - Standard Barbed Wire Fence
- Type D - Chain Link Fence.

Chain link fence protection on rock slopes and tentative requirements for High-tensile Smooth-wire Fences are also included.

316.02 Applicable Documents - See Table 316-A.

Note: Standards referred to in this Section are obtainable from:

CGSB: Canadian Government Publishing Centre
Supply and Services Canada
Ottawa, ON K1A 0S9

ASTM: American Society for Testing and Materials
1916 Race Street
Philadelphia, PA 19103

316.03 Type A & B Fence Fabrics - Fence fabric for railway and highway rights-of-way shall be manufactured from good commercial quality galvanized steel wire in general accordance with ASTM A 116, but shall conform to the requirements following and those listed in Table 316-B.

316.03.01 Styles - See Table 316-B. Length of fabric rolls shall be 101 m unless otherwise ordered.

Note: Other standard and specialty fabrics when specified shall meet the following general specifications.

316.03.02 Wire Size and Strength - All wire used in the fabric shall have a minimum galvanized coated nominal diameter of 2.50 mm (12½ ga.). The variation permitted from the measured nominal diameter shall be ± 0.1 mm. The minimum breaking force shall be 4250 N.

316.03.03 Zinc Coating - The zinc used for coating shall be any grade of zinc conforming to ASTM B 6. All wire shall have a uniform coating of zinc which shall adhere firmly to the base metal. The average mass of zinc coating shall be 90 g/m^2 of uncoated wire surface for three specimens but minimum 80 g/m^2 for any individual specimen.

316.03.04 Quality of Work - Splicing of individual line wires by means of a wrap joint, electric butt weld or galvanized sleeve is permitted.

The maximum number of line wire joints shall not exceed half of the number of line wires in any roll of fencing.

All joints shall be made in accordance with good work practice.

All stay wires shall be properly spaced and substantially perpendicular to the line wires.

The locks or fastenings at the intersections of the longitudinal and vertical wires of the woven wire fencing shall be of such construction as will prevent them from slipping, either longitudinally or vertically. All locks and fastenings shall be effectively galvanized.

Horizontal wires of all woven wire fencing shall be provided with tension curves to accommodate temperature changes after erection.

316.03.05 Testing

i) Samples: Sufficient length of fabric to include at least three vertical stay wires shall be cut from each roll to be tested as required by DBSS 316.15 from which line wire and stay wire specimens shall be taken for testing.

ii) Diameter of coated wire: The average of two readings taken at right angles to each other at random on uniformly coated areas shall all be within the tolerance measurements specified in DBSS 316.03.02.

iii) Strength: The tensile strength of three line wire specimens shall be tested in accordance with ASTM A 370 on a machine calibrated to ASTM E 4 standards. The tensile strengths shall meet the requirements of DBSS 316.03.02.

iv) Coating adhesion: Three line wire specimens shall be wrapped in a close helix at least twice around a cylindrical mandrel having a diameter four times the nominal diameter of the wire: wrapping speed shall not exceed four seconds per turn. The coating shall adhere to the wire without flaking or being removed by rubbing with bare fingers.

v) Coating mass: Three line wire specimens (from each roll), each more than 400 mm in length, and three stay wire specimens of short lengths between knots, etc., but each specimen totalling more than 400 mm shall be tested in accordance with ASTM A 90 for conformity with the requirements of DBSS 316.03.03.

vi) Rejection and Retest: Refer to DBSS 316.16.

316.04 Type D Fence Chain Link Fabric - Chain link fence fabric shall be manufactured from good commercial quality galvanized steel wire (or other specified finish) in general accordance with CAN/CGSB-138.1M and shall conform to the requirements listed in DBSS 316.04.01 through DBSS 316.04.04.

316.04.01 Fabric Classification (according to CGSB Standard) - The type, class, style and grade of chain link fence fabric shall be as specified in Special Provisions in accordance with the Drawings and/or Drawings SS SP741-05.01 and SS SP741-05.02, selected from the following:

i) Type/Class 1A - Zinc coated (galvanized) steel wire fabric normally will be specified for all heights of chain link fences as indicated on Drawing SS SP741-05.01, unless Type/Class 1B Vinyl coated steel is required for authorized special installations.

ii) Style - nominal diameter of zinc coated wire:

- Heavy, for highway protection fence from rolling rock
- Medium, for Standard and Security height fences
- Light, for Residential height fences (see Drawing SS SP741-05.01)
- See Table 316-C and 316-D (when permitted).

iii) Grade - mass of zinc coating per unit area of uncoated wire: Grade 1, average 490 g/m^2 for three specimens, minimum 440 g/m^2 for any individual specimen. The zinc used for coating shall be any grade of zinc conforming to ASTM B 6. All wire shall have a uniform coating of zinc which shall adhere firmly to the base material.

Note: When specified, Grade 2 heavy coating (average 610 g/m^2, individual 550 g/m^2 minimums) for corrosive environments and the zinc coating (average 90 g/m^2, individual 80 g/m^2 minimums) for vinyl coated wire and the vinyl coating requirements shall be in accordance with CAN/CGSB-138.1M.

316.04.02 Fabric Height, Length and Mesh - Wire shall be woven throughout to a uniform diamond mesh 50 mm ± 3.0 mm clear between parallel mesh wires: fabric height as specified ± 25 mm with both selvage edges twisted and/or knuckled:

- fabric 1.5 m high and under: knuckled both edges
- fabric 1.8 m high and over: knuckled one edge, and twisted the other
- Residential height fence light style fabric 1.0 m and 1.2 m high: knuckled both edges.

Length of rolls with slack removed by nominal tensioning shall be 100 m unless otherwise ordered.

Note: Typical (normal) ordering designation from above: Type 1, Class A, Grade 1, Medium Style with roll width (height), length, mesh size and quantity.

316.04.03 Quality of Work - Fabric shall be of uniform quality and condition, free from all defects and imperfections affecting serviceability and appearance.

316.04.04 Testing

i) Samples: Three full height pickets shall be provided from each roll as required by DBSS 316.15.

ii) Diameter of coated wire: The average of two readings taken at right angles to each other at random on uniformly coated portions of each pair of pickets between bends shall be within the tolerance measurement specified in DBSS 316.04.01(ii).

iii) Strength: The tensile strength of three line wire specimens shall be tested in accordance to ASTM A 370 on a machine calibrated to ASTM E 4 standards. The tension applied to the undeformed portion between two adjacent bends shall meet the requirements of DBSS 316.04.01(ii).

iv) Coating adhesion: Three specimens of undeformed portions between two adjacent bends shall be wrapped in a close helix once around a cylindrical mandrel having a diameter four times

TABLE 316-A DOCUMENTS APPLICABLE FOR REFERENCE

CAN/CGSB-138.1-M and 1-GP-181M	Fence, Chain Link, Fabric Coating, Zinc-Rich, Organic, Ready Mixed
ASTM A 53	Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
ASTM A 90	Test Method for Weight of Coating on Zinc-Coated (Galvanized) Iron or Steel Articles
ASTM A 116	Specification for Zinc-Coated (Galvanized) Steel Woven Wire Fence Fabric
ASTM A 121	Specification for Zinc-Coated (Galvanized) Steel Barbed Wire
ASTM A 123	Specification for Zinc (Hot-Galvanized) Coatings on Products Fabricated from Rolled, Pressed, and Forged Steel
ASTM A 153	Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A 641	Specification for Zinc-Coated (Galvanized) Carbon Steel Wire
ASTM B 6	Specification for Zinc (Slab Zinc)
ASTM B 211M	Specification for Aluminum-Alloy Bar, Rod, and Wire [Metric]

TABLE 316-B STYLES OF FENCES

Fence Types	Fabric Style No.	Number of Horizontal Wires	Fabric Height (mm)	Stay Wire Spacing (mm)	Wire Diameter (mm)
A	1047-6	10	1195	150	2.50 MIN
B	939-6	9	990	150	2.50 MIN

TABLE 316-C ZINC COATED WIRE

Type 1A	Style	Nominal Diameter of Zinc Coated Wire	Breaking Force Minimum N
	Heavy	5.0 mm ± 0.1 mm	10 000
	Medium	3.5 mm ± 0.1 mm	5 000
	Light	3.0 mm ± 0.1 mm	3 600

the nominal diameter of the wire: wrapping speed shall not exceed four seconds per turn The coating shall adhere to the wire without flaking or being removed by rubbing with bare fingers.

v) Coating mass: Three specimens (from each roll) each more than 400 mm long excluding knuckled or twisted ends shall be tested in accordance with ASTM A 90 for conformity with the requirements of DBSS 316.04.01 (iii).

vi) Rejection and Retest: refer to DBSS 316.16.

316.05 Chain Link Fabric and Accessories on Rock Slopes - Chain link fabric for protection on rock slopes shall be manufactured and tested in general conformity with DBSS 316.04 and to the following requirements:

- Fabric shall be to CAN/CGSB-138.1M designation of wire woven to 75 mm diamond mesh in rolls 3.0 m wide, knuckled on both selvage edges, and 15 m long. See Table 316-E.

Note: For corrosive conditions and environment Grade 2 - heavy galvanizing may be specified as a special order with possible extended lead time required for delivery.

316.06 Barbed Wire - Barbed wire for Type B & C fences and for security height Type D fences shall be manufactured from good quality galvanized steel wire in general accordance with ASTM A 121 but shall conform to the following detail requirements:

a) Wire Size and Strength: The wire shall be continuously twisted double strand of 2.5 mm nominal diameter (12 1/2 ga.) and to have a minimum breaking force of 4000 N.

b) Barbs: Four point barbs shall be of not less than 2.0 mm nominal diameter (14 ga.) and regular 150 mm maximum spacings.

c) Zinc Coating Class (according to ASTM A 121): The zinc used for coating shall be any grade of zinc conforming to ASTM B 6. All wire, including barbs, shall have a uniform coating of zinc which shall adhere firmly to the base metal.

The average mass of zinc coating per unit area of uncoated wire for three specimens and the minimum for any individual specimens shall be in accordance with the class of coating as follows:

i) Class 1 for use in the construction of Type C fences and for the top strand of Type B fences (see Drawings SS SP741-01.01 and SS SP741-02.01), the coating shall be average 90 g/m^2 and 80 g/m^2 individual minimums.

ii) Class 3 for use with Security height Type D Chain Link fences (see Drawing SS SP741-05.01), coating shall be average 240 g/m^2 and 215 g/m^2 individual minimums.

d) Length: Spools shall be 400 m unless otherwise ordered.

e) Quality of Work: Splicing of individual strand wires by means of a wrap joint or an electric butt weld is permitted. Not more than three splices or joints shall exist in any 400 m spool of barbed wire. Such splices or joints shall be made in accordance with good work practice.

The twist in the double strand barbed wire shall be uniform and continuous in one direction to lock the barbs in place. The barbs shall be sharp and well formed.

TABLE 316-D VINYL COATED ZINC-COATED WIRE (when permitted)

Type 1B	Style	Vinyl Coated Zinc-Coated Wire Nominal Diameter	Breaking Force Minimum N
	Heavy	5.0 mm ± 0.1 mm	6 000
	Medium	3.5 mm ± 0.1 mm	3 000
	Light	3.0 mm ± 0.1 mm	2 600

TABLE 316-E FABRIC DESIGNATION

i.e.	**Type 1**	**Class A Grade 1**	**Light Style**	**OR** Nominal Diameter	**Medium Style**
	Steel	zinc coated (galvanized)	3.0 mm		3.5 mm

Wire strands showing nicks or indentures likely to cause the wire to break under flexure are not acceptable.

f) Testing

i) Samples: Sufficient length of barbed wire shall be cut from each spool to be tested as required by DBSS 316.15.

ii) Strength: The tensile strength of three line wire specimens shall be tested in accordance to ASTM A 370 on a machine calibrated to ASTM E 4 standards. The tensile strength shall meet the requirements of DBSS 316.06 (a).

iii) Coating Mass: Three single wire specimens (from each spool), each aggregating more than 400 mm in length shall be tested in accordance with ASTM A 90 for conformity with the requirements of DBSS 316.06 (c).

iv) Rejection and Retest: Refer to DBSS 316.16.

Note: Smooth (barbless) galvanized wire, not specified to be "high-tensile," shall be double strand meeting all the above requirements except those for barbs.

316.07 High-tensile Wire - Requirements for galvanized wire for high-tensile smooth-wire (no barbs) fencing are shown in Table 316-F.

316.08 Brace, Tension and Tie Wire

316.08.01 Brace Wires - Brace wire shall be manufactured of good commercial quality soft temper steel wire 3.5 mm nominal diameter gauge and zinc coated (galvanized) normally minimum 80 g/m^2 (Class 1 according to ASTM A 641) or as otherwise specified or approved.

Brace wire for high-tensile smooth-wire fencing may be fence wire specified in DBSS 316.07.

316.08.02 Tension Wire - Tension wire to secure bottom of chain link fencing and at top where specified shall be manufactured of good commercial quality coil spring crimped steel wire 5.0 mm nominal diameter gauge and zinc-coated (galvanized) normally 400 g/m^2 (Class B according to ASTM A 641) for Standard and Security height fences, or as otherwise specified or approved. Residential height fences tension wire shall be 3.5 mm nominal diameter.

316.08.03 Tie Wire and Wire Hog Ring Clips - Tie wire and wire hog ring clips shall be of steel of the nominal diameter gauge specified and zinc coated equivalent to that specified for the relevant fencing fabric: Aluminum tie wire shall be of specified gauge and according to ASTM B 211M.

316.09 Steel Fence Posts on Rock (Excluding Chain Link Fence) - Steel fence posts and braces for Type A, B and C fences where specified or required on exposed rock or rock with minimum over-burden shall be of good commercial quality steel conforming to either Category A or Category B as specified in DBSS 316.10 with the following exceptions:

- Line post mass shall be not less than 1.9 kg/m and angles 4.75 kg/m minimum, both drilled for

TABLE 316-F REQUIREMENTS FOR GALVANIZED WIRE FOR HIGH TENSILE SMOOTH-WIRE (NO BARBS) FENCING

Diameter:	2.5 mm nominal diameter (12 1/2 ga.)
Tensile Strength:	between 1170 MPa and 1590 MPa (170,000 to 230,000 psi)
Breaking Force:	in excess of 5800 N (1300 lbf)
Zinc Coating (Class 3):	minimum 240 g/m^2
Coil Length:	in excess of 1200 m (approx. 100 lb.)

TABLE 316-G REQUIREMENTS FOR PIPE COMPONENTS FOR CHAIN LINK FENCING - CATEGORY A

Pipe Sizes	Nominal Outside Diameter (mm)		
	Residential	Standard	Security
Line Posts	42	48	60
End, Corner & Straining Posts	48	60	73
Gate Posts, gates 2.5 m and less	48	60	73
Gate Posts, gates over 2.5 m	48	73	89
Top Rails	33	33	42
Brace Rails	--	33	42

attaching barbed wire or a substantial number of fence fabric line wires with wire ties or clips.

- Channel braces shall be 1.5 kg/m minimum.

316.10 Steel Pipe Posts, Rails and Braces - Chain Link Fencing

316.10.01 Category and Sizes - All posts, rails and braces shall be of Category 1 or Category 2 at the discretion of the supplier and shall be supplied in the nominal outside diameter as specified in Special Provisions or in accordance with the Drawings and/or Drawing SS SP741-05.02 and selected from Tables 316-G or 316-H or as may be otherwise specified.

Note: Specifications may permit thin wall tubing and consequent lower zinc coating for proprietary galvanized and vinyl coated residential fence posts and gates.

Bolts and nuts for fastening braces to posts shall be galvanized Class D (305 g/m^2 individual pipe posts, rails and braces for chain link Type D fences shall be manufactured in general accordance with ASTM A 53.

a) Category A - Category A posts, braces and rails shall be steel butt weld pipe, Schedule 40, as specified by ASTM A 53, except that the hydrostatic testing requirement is waived, and the minimum weight shall be no less than 95% of the nominal weight. Pipe shall be zinc-coated in accordance with ASTM A 53. This ASTM Standard requires, among other things, an average zinc coating of 550 g/m^2 and a minimum zinc coating of 490 g/m^2.

b) Category B - Category B posts, braces and rails shall be made from steel pipes. The pipe shall be formed from steel strip conforming to ASTM A 569. The steel in the formed pipe shall have a minimum yield strength of 350 MPa (50,000 psi). The product of the yield strength and section modulus of the pipe shall be not less than that of pipe meeting the requirements for Category A posts and rails.

The outer surface of Category B posts, braces and rails shall have all of the following protective coatings, applied in the following order:

i) Hot dip zinc coating, conforming to ASTM B 6, High Grade or Special High Grade, applied

TABLE 316-H REQUIREMENTS FOR PIPE COMPONENTS FOR CHAIN LINK FENCING - CATEGORY A

Actual Outside Post Diameter (mm)	26.7	33.4	42.1	48.3	60.3	73.0	88.9
Wall Thickness (mm)	2.87	3.38	3.56	3.68	3.91	5.16	5.49
Mass (kg/m)	1.7	2.5	3.4	4.0	5.4	8.6	11.3
Tolerances:	Diameter ± 1%			Mass ± 5%			

at an average weight of 305 g/m^2 (1.0 oz/ft^2) and at a minimum weight of 275 g/m^2 (0.9 oz/ft^2), as determined by the test method in ASTM A 90.

ii) Chromate conversion coating, with a weight of 30 micrograms/sq. in.
± 15 micrograms/sq.in., as determined by a strip and weight method for the zinc and chromate and by determining the percentage of each by atomic absorption spectrophotometer.

iii) Clear acrylic polyurethane coating with a thickness of 0.100 mm ± 0.025 mm (0.4 ± 0.1 mils).
The combined outer coating shall have an ability to resist:

- 1000 hours of exposure to salt fog with a maximum of 5% red rust when tested in accordance with ASTM B 117.
- 500 hours of exposure to 100% relative humidity without blistering and peeling when tested in accordance with ASTM D 2247; and
- 500 hours of exposure in a weatherometer without film cracking of the clear coat when tested in accordance with ASTM G 23, Type E or EH carbon arc, or ASTM G 26, Type B or BH xenon arc.

The inner surface of Category B, posts, braces and rails shall have either a zinc coating as per ASTM F 1234, Type B, applied at an average rate of 305 g/m^2 and a minimum rate of 275 g/m^2; or an ASTM F 1234, Type D zinc coating with a minimum zinc powder loading of 91% by weight and a minimum thickness of 0.075 mm (0.3 mils). The inner coating shall have an ability to resist 650 hours of exposure to salt fog with a maximum 5% red rust when tested in accordance with ASTM B 117.

316.10.02 Quality of Work - The finished pipes shall be reasonably free from injurious defects. All burrs at ends of pipes shall be removed.

The zinc or zinc/chromate/acrylic coating shall be free from injurious defects and excessive roughness.

316.10.03 Testing

a) Samples: A 100 mm specimen shall be cut from each end of one pipe in every lot of 500 or fraction thereof.

Note: Specimens may be taken from the extreme ends of pipe that has been cut to length after galvanizing; otherwise, specimens shall be taken at least 200 mm clear of each galvanized end build-up.

b) Dimensions: Specimens shall be measured for diameter and wall thickness/mass and all shall be within the tolerance measurements specified by DBSS 316.10.01.

c) Mass Coating: Specimens shall be tested for the mass of zinc or composite coating in accordance with DBSS 316.10.01 and ASTM A 90.

d) Rejection and Retest: Refer to DBSS 316.16.

316.11 Gates and Hardware - Gates shall be supplied with hardware ready for installation as specified by the Purchase Order, Work Order or Special Provisions in accordance with the Drawings and/or Drawings SS SP741-04.04, SS SP741-04.05, SS SP741-05.03 and SS SP741-05.04 or as may be otherwise specified or ordered.

Gates may be specified by the types as follows:

- **Type I** Pressed sheet steel frame, in-fill and brace members
- **Type II** Pipe frame and horizontal pipe in-fill members
- **Type III** Pipe frame, bracing and wire fabric in-fill
- **Type IV** Pipe frame, bracing and chain link fabric in-fill
- **Type V** Slip wire range gates barbed wire and wood droppers

with the general requirements of material, finish, hardware and quality of work specified in DBSS 316.11.01 through DBSS 316.11.06.

316.11.01 General - Gate sizes shall be the fence height or as otherwise stipulated and the nominal width designated by the clear opening width between gate posts with allowance made for hinges and latches, etc.

Gates shall be fabricated of galvanized pipe, tubing or formed sheet metal sections properly joined, braced and with in-fill all as specified by type below.

All pipe joints shall be cut to fit closely together and fully welded, vertical stiles neatly capped or mitred to horizontals with drilled holes to vent and drain closed

end pipe members. Bends where permitted at corners shall be made without deforming tube or pipe section.

All welds, conforming to the best commercial practice, shall be smooth, and with adjoining and other marred galvanized pipe surfaces, shall be painted with two heavy coats of zinc rich paint manufactured in accordance with Standard 1-GP-181M.

All hardware and required fittings and fasteners shall be of suitable aluminum alloy or of steel or malleable or ductile cast iron, galvanized in accordance with ASTM A 153. Hinges generally shall be so designed as to permit the gate to swing back against the fence and to prevent the easy removal of gate and hinges. Hinges and other fittings shall all be of adequate strength for fastening securely without distortion and shall be sound and free of defects. Chain and/or latching gate catch for padlocking together with keepers to hold the gate open and centre drop bolt and sleeve for double swing gates shall be provided as and where specified or ordered.

All galvanizing and other finished surfaces shall be smooth and free of sharp dangerous protrusions.

316.11.02 Type I Gate (sheet metal members) - Formed heavy duty galvanized sheet metal sections with rolled edges as horizontal rails and vertical stiles, braced vertically and/or diagonally with similar members as necessary shall be rigidly riveted together to form a rugged non-sagging prefabricated gate.

Nominal sizes: 1220, 2440, 3050, 3660, 4270 and 4880 mm wide and 1220 mm high.

Gates to be supplied completely assembled with required hardware for wood (or metal) gate posts all as specified or ordered.

316.11.03 Type II Gates (all pipe) - Galvanized pipe 42 mm OD, 3.4 kg/m mass with zinc coating minimum 490 g/m^2 (see DBSS 316.10.01 and DBSS 316.10.03) shall be welded together as gate frame and horizontal pipe in-fill members at 305 mm on centres. Vertical centre bracing of 5 mm x 19 mm flat bar on both sides of all horizontals is not required for gates 3050 mm and 3660 mm nominal widths. Gussets shall be welded to opposite corners of the gate as shown on Drawing SS SP741-04.05.

Nominal sizes: 3050, 3660, 4270 and 4880 mm wide and 1220 mm high.

Gates shall be supplied completely assembled with required hardware for wood (or metal) gate posts all as specified or ordered.

316.11.04 Type III Gates (pipe and fabric) - Galvanized pipe 33 mm OD, 2.5 kg/m mass with zinc coating minimum 490 g/m^2 (see DBSS 316.10.01 and DBSS 316.10.03) shall be formed with even radius bent corners or neatly welded corners and vertical bracing at 1/3 points in the width of 27 mm OD, 1.7 kg/m pipe. Cleats or gussets shall be welded to opposite corners of gate frame/panel(s) as indicated on Drawing SS SP741-04.05, holed for securing under tension a diagonal brace consisting of 5.0 mm nominal diameter galvanized coil spring wire (see DBSS 316.08) with or without forged turn-buckle as and when specified.

In-fill shall be wire fabric all of 3.5 mm galvanized diameter wire with seven or more horizontal line wires, and stay wires not more than 600 mm apart. Fabric shall be securely fixed taut without sag to gate frame all round and to vertical brace.

Nominal sizes: 3050, 3660, 4270, 4880 and 5490 mm wide and 1070 and 1220 mm high.

Gates shall be supplied completely assembled with required hardware for wood (or metal) gate posts all as specified.

316.11.05 Type IV Gates (pipe and chain link) - Gate frame and required vertical bracing and/or diagonal bracing to cleats all welded as generally indicated on Drawing SS SP741-05.04 for hinged swing gates shall be fabricated to the designated size(s) complete with chain link in-fill and hardware for fixing to metal posts all as specified.

Pipe shall be in accordance with DBSS 316.10 and welding with DBSS 316.11.01. In-fill of chain link fabric as specified for fencing (see DBSS 316.04), shall be attached to gate frame stiles with tension bars and bands (see DBSS 316.12), and to horizontal members and vertical bracing with tie wire (see DBSS 316.08.03), all taut and sag free.

Sliding gates may be specified for gate openings over 4.0 m and shall be fabricated as generally indicated for swing gates but with diagonal tension rod bracing to support centre from both ends. The rolling hardware shall hold the gate vertical and parallel to the gate posts and adjoining fence at all times to allow free operation from any position while not retained by latches or keepers.

Roller assembly sheaves and wheels with bearings, grease fittings, necessary clamps, supports and fasteners, and chain and/or latching gate catch for padlocking, all approved by the designer and shall be provided as and when specified.

316.11.06 Type V Gates (range slip wire) - Barbed wire continued across standard width gate opening (for possible future farm gate) with 50 mm diameter full fence height treated wood droppers securely figure of eight wire tied and stapled to each line wire shall be provided with a galvanized steel strap loop at bottom and either similar loop at top with an intermediate chain (for padlocking) or, when specified, a Princeton or other approved cinch-up lockable latch device all for securely attaching to fence or gate posts to hold open-end dropper.

316.12 Accessory Material - All required fittings, fasteners, hardware and accessories shall be of suitable aluminum alloy or steel, malleable or ductile cast iron, and galvanized in accordance with ASTM A 153, or as may be specified.

a) Staples for securing wire and fabric to treated wood fence posts shall be nominal 3.5 mm diameter steel wire with a driving length of at least 45 mm heavy galvanized finish.

b) Hog ring clips for securing wire fabric and/or barbed wire of Type A, B and C Fences to steel posts and to tension wire of Type D Fences see DBSS 316.08.03.

c) Dowel, Spikes, Bolts and Nuts shall be of good commercial quality, galvanized where specified.

d) Metal Droppers for high-tensile fencing will be prefabricated "Waratah" or other acceptable galvanized formed sheet metal droppers, 1070 mm for 5-wire and 1170 mm long for 6-wire fences, with preformed wire clips: such metal droppers or other acceptable manufacture may be specified or permitted as an alternative to the wood droppers specified by DBSS 909.08 for barbed wire (Type C) fences.

Note: Galvanized twisted double wire fence stays are permitted only for the extension of normal fence heights to deer height and the like.

e) All terminal and tensioning posts, for Type D chain link fences, shall be provided with aluminum or galvanized weather tight caps to fit and fasten securely and with extended arms for engaging barbed wire of security fences.

f) Caps for line posts for Type D chain link fences, shall carry top rail. Rail and brace end caps shall be provided for securing to rails and bolting with steel bands to posts.

g) Tension Bars for securing chain link fencing to terminal and tensioning posts and to vertical stiles of gates of Type D Fences shall be merchant quality mild steel strip 5 mm x 19 mm in minimum lengths of 50 mm less than full height of the chain link fabric, hot dip galvanized.

h) Tension and Brace Bands for securing brace and top rail caps and tension bars to posts shall be formed of minimum 2.5 mm x 19 mm mild steel strip, hot dip galvanized complete with bolt and nut.

i) Tensioning Devices for individual in-line high-tensile wire will be "Hayes" or other acceptable manufacture and operation.

j) Mechanical Splices and Fasteners shall be of "Nicopress FW2-3" or other acceptable manufacture applicable to the respective diameter and tensile strengths of wires.

316.13 Packaging and Marking - Each roll of fencing fabric or chain link mesh and spool of wire shall be tightly rolled and firmly tied to prevent loosening during shipment and handling.

Posts shall be securely fastened in bundles or sets, and accessories packaged in cartons or other suitable containers, to prevent loss or damage during shipment and handling.

Each product, roll, bundle or container etc., shall be clearly marked with the name, brand and/or trade mark of the manufacturer; the heat or lot number and other information sufficient to identify the style, type, class, grade, finish, height and/or length, nominal or outside diameter, gauge and the applicable Standard to which the product conforms.

316.14 Inspection - No material shall be shipped prior to inspection or until a release for shipment has been issued by the Contractor.

All material may be inspected before shipment to the construction site or at the construction site, or both. The Supplier or manufacturer shall provide the necessary facilities to enable the Contractor and the

Ministry to expeditiously examine selected pieces from the material lot(s) clearly identified for intended supply to the Contractor. Identification shall be such that after inspection and testing the lot(s) may be subsequently identified for acceptance or rejection.

Inspection of material before shipment shall not exclude its subsequent rejection at the construction site if found to fail any requirements of this specification.

316.15 Sampling - One roll of fabric or spool of wire, in every lot of 50 or fraction thereof, in the shipment may be selected at random for testing but in no case will less than two samples be tested.

Specimens for testing will be required from the outside ends of the selected sample roll or spool material and both ends of one post in every 500 or fraction thereof.

Sufficient length of each roll, spool or pipe shall be taken to provide the number of specimens for each of the testing requirements specified.

Wire used for knots, knuckling, twisting and wraps and wire containing welds shall be excluded from tests.

See respective material tests for pertinent sample lengths.

316.16 Rejection and Retests - If any tested specimen fails to meet the testing requirements specified above, the roll, spool or pipe represented by the specimens shall be replaced with two additional rolls, spools or pipes randomly selected by the Quality Manager and both shall meet the test requirements in every respect, otherwise the complete lot represented by the samples will be subject to rejection.

In the event of complete lot rejection and prior to any further testing by the Ministry or the Contractor, the intended replacement lot(s) shall be satisfactorily tested by a mutually acceptable independent testing agency at the expense of the Contractor. Where the time delay for retesting is unacceptable to the Contractor the order or part thereof will be subject to cancellation.

Rejected material shall be replaced at the Supplier's or Contractor's expense including shipping charges and removal of rejected material at the construction site, if applicable.

SECTION 317

P.E. PLASTIC DRAINAGE PIPE

317.01 Scope - This Section covers the requirements and methods of testing corrugated polyethylene (P.E.) drainage pipe. The inside wall may be corrugated or smooth. It covers nominal pipe diameters 200 mm, 250 mm, 300 mm, 400 mm, 450 mm and 600 mm.

317.02 Applicable Documents

ASTM D 883: Definition of Terms Relating to Plastics

ASTM D 2122: Standard Method of Determining Dimensions of Thermoplastic Pipe and Fittings

ASTM D 2412: Test for External Loading Properties of Plastic Pipe by Parallel Plate Loading

ASTM D 2444: Standard Test Method for Impact Resistance of Thermoplastic Pipe and Fittings by means of a Tup (Falling Weight)

ASTM F 405: Standard Specification for Corrugated Polyethylene Tubing and Fittings

ASTM F 412: Definition of Terms Relating to Plastic Piping Systems

ASTM F 667: Standard Specification for 8", 10", 12", 15", 18" and 24" Corrugated Polyethylene Tubing

Note: ASTM Standards are obtainable from:
American Society for Testing and Materials,
1916 Race Street,
Philadelphia, PA 19103.

317.03 Materials - The pipe, fittings and couplers shall be manufactured from virgin polyethylene plastic of Type III or IV, Class C, Category 5 and Grade P33 as defined in ASTM D 1248.

317.04 Joining Systems - The assembly of joints shall be in accordance with the pipe manufacturer's recommendations.

Approved joint systems are as follows:

a) Integral Bell Gasketed Joint

i) The joint shall be designed so that when assembled, the gasket will be compressed radially on the pipe spigot or in the bell to form a watertight seal.

ii) The joint shall be designed to avoid displacement of the gasket when assembled in accordance with the manufacturer's recommendation.

b) Corrugated Screw-On Coupler (Nominal Pipe Diameters 250 and Larger)

i) Coupler shall be manufactured of polyethylene plastic as specified in DB SS 317.03.

ii) Coupler length shall be a minimum of 1.1 times the nominal pipe diameter or 270 mm, whichever is greatest (a minimum of 0.4 times the length of the coupler shall be engaged on both pipes to be joined).

iii) Average coupler wall thickness shall be a minimum 2.6 mm for nominal pipe diameters of 250 mm and 300 mm and 3.2 mm for all other diameters up to and including 600 mm.

317.05 Requirements

317.05.01 Quality of Work - The pipe shall be homogenous throughout and free from visible cracks, flaws, foreign inclusions or other injurious defects. The pipe shall be uniform in colour, opacity and other physical properties.

317.05.02 Pipe Diameter

i) Nominal Diameter - The nominal diameter shall be the average inside diameter rounded to the nearest whole number (mm).

ii) Inside Diameter - The permissible inside diameter tolerances shall be +3% and -1.5%. Measurement shall be taken in accordance with ASTM D 2122.

317.05.03 Length - Laying length shall be 4 m or 6 m for all diameters otherwise specified on the Design. A tolerance of ± 25 mm on the nominal laying length will be permitted.

317.05.04 Physical Properties

i) Impact Strength - The minimum drop weight impact strength shall be 36.8 Joules when tested in accordance with DB SS 317.07.01.

ii) Pipe Stiffness - The minimum pipe stiffness shall be 210 kPa when tested in accordance with DB SS 317.07.02.

iii) Pipe Flattening - There shall be no evidence of splitting, cracking or breaking when tested in accordance with DB SS 317.07.03.

317.05.05 Joint Separation Resistance - Joint systems when installed in accordance with the manufacturer's instructions shall not separate when tested in accordance with DB SS 317.07.04.

317.06 Marking - All pipe shall be clearly marked at intervals of no more than 1.5 m with 5 mm or larger letters with the following information:

- Manufacturer's name or trademark
- Nominal diameter
- Materials designation and cell class: PEGR33
- The words "Drainage" or similar
- The class: PS210
- The applicable Specification designation (ASTM F 405 or F 667)
- Date of manufacture and plant designation.

317.07 Test Methods

317.07.01 Impact Resistance - Five specimens, each at least as long as the nominal pipe diameter, shall be tested in accordance with ASTM D 2444, using a 2.5 kg Tup "B" and the flat plate holder "B". The height of the drop shall be 1.5 m (total drop energy = 36.75 Joules). The test shall be performed at 0°C (±0.5°) allowing no more than 15 seconds to complete the test after removing the sample from the conditioning medium. Any cracking or other signs of material fatigue evident after the test shall constitute a failure. All five specimens must pass. If one specimen fails, five more specimens shall be tested. Nine out of ten specimens passing will be acceptable.

317.07.02 Pipe Stiffness - The pipe stiffness shall be determined at 5% deflection of the initial inside diameter in accordance with ASTM D 2412. Three specimens, each at least as long as the nominal pipe diameter, shall be tested. The exact specimen length shall be an integer multiple of the corrugation pitch. All test specimens shall meet the required stiffness.

317.07.03 Pipe Flattening - The three specimens tested in DB SS 317.07.02 shall be flattened between parallel plates in a suitable press until the distance between the plates is 40% of the original inside diameter. The loading rate shall be uniform and such that the compression is completed within two to five minutes. Remove the load and examine the specimen. All three specimens shall meet the requirements.

317.07.04 Joint Separation - A test specimen shall be constructed using the manufacturer's joining system centred on two pieces of pipe each 0.75 m in length. The specimen shall be hung vertically and a dead weight test force applied to the bottom end of the pipe. The vertical test force shall be 0.87 N/mm of nominal pipe diameter. Any separation of the joint shall be considered a failure.

317.08 Inspection

a) The Quality Manager and Ministry shall have access to the fabrication plant for inspection and every facility shall be extended for this purpose.

b) Inspection may include checks on physical dimensions and general quality of work.

c) For the purpose of sampling for DB SS 317.05 and 317.07, one pipe from each lot of 50 pipes or fraction thereof in a shipment shall be selected for sampling. All test specimens shall be cut from the selected pipe(s).

d) If the test sample fails to meet the requirements of DBSS 317.05 when tested in accordance with DB SS 317.07, that batch will be rejected and shall be replaced by the Supplier with a new batch that conforms to this Specification.

317.09 Quality Assurance

a) The Supplier shall develop and maintain an effective quality control system to ensure that adequate inspection coverage is maintained throughout the manufacturing process. Evidence of such inspection shall be available to the Quality Manager and the Ministry prior to shipment from the place of manufacture or the Supplier's storage facility.

b) The Contractor shall, and the Ministry reserve the right to, audit the Supplier's quality control system to ensure conformance with this Specification.

SECTION 318

PVC PLASTIC DRAINAGE PIPE

318.01 Scope - This Section covers the requirements and methods for testing for:

- Type PSM* Poly Vinyl Chloride Drainage Pipe
- Ribbed Poly Vinyl Chloride Drainage Pipe

and the perforation requirements of type PSM pipe up to a nominal diameter of 375 mm.

Note: The term PSM* is not an abbreviation but rather an arbitrary designation for a product having certain outside dimensions.

318.02 Applicable Documents

CSA B182.2: Large-diameter, Type PSM PVC Sewer Pipe and Fittings

CSA B182.4: Large-diameter, Ribbed PVC Sewer Pipe and Fittings

ASTM D 883: Definition of Terms Relating to Plastics

ASTM D 1784: Standard Specification for PVC and CPVC Compounds

ASTM D 2122: Standard Method of Determining Dimensions of Thermoplastic Pipe and Fittings

ASTM D 2152: Standard Test Method for Quality of Extruded PVC and CPVC Compounds

ASTM D 2412: Test for External Loading Properties of Plastic Pipe by Parallel-Plate Loading

ASTM D 2444: Standard Test Method for Impact Resistance of Thermo-plastic Pipe and Fittings by Means of a Tup (Falling Weight)

ASTM D 3034: Standard Specification for Type PSM PVC Sewer Pipe and Fittings

ASTM D 3212: Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals

ASTM F 412: Definition of Terms Relating to Plastic Piping Systems

ASTM F 477: Standard Specification for Elastomeric Seals for Joining Plastic Pipe

ASTM F 794: Standard Specification for PVC Large-diameter Ribbed Gravity Sewer Pipe and Fittings Based on Controlled Inside Diameter.

318.03 Materials - The pipe shall be made of virgin PVC plastic having a cell classification of 12454-B, 12454-C, or 13364-C, as defined in ASTM D 1784.

The fittings shall be made of virgin PVC plastic and shall have a cell classification of 12454-B, 12454-C, or 13343-C, as defined in ASTM D 1784.

Elastomeric gaskets shall comply with the requirements of ASTM F 477.

318.04 Quality of Work - The pipe shall be homogenous throughout and free from visible cracks, flaws, foreign inclusion, or other injurious defects. The pipe shall be uniform in colour, opacity and other physical properties.

318.05 Joining System - All sizes of pipe shall be supplied with Elastomeric Gasket Joints providing a watertight seal in accordance with ASTM D 3034.

a) Integral Bell Gasketted Joints shall be designed so that when assembled, the gasket shall be compressed radially on the pipe spigot or in the bell to form a watertight seal.

b) The joint shall be designed to avoid displacement of the gasket, when assembled in accordance with the manufacturer's recommendation.

c) Joint assembly shall be in accordance with the pipe manufacturer's recommendations.

318.06 Requirements

318.06.01 Length - Laying length shall be 4 m or 6 m for all diameters unless otherwise specified by the Design. A tolerance of ± 25 mm on the nominal laying length will be permitted.

318.06.02 Type PSM, PVC Drainage Pipe Dimensions - The average outside diameter and the minimum wall thickness of Type PSM PVC drainage pipe shall be as specified in Table 318-A when measured in accordance with ASTM D 2122.

TABLE 318-A PIPE DIMENSIONS - TYPE PSM

OUTSIDE DIAMETER (mm)					
NOMINAL DIAMETER (mm)	AVERAGE	TOLERANCE	MINIMUM	MAXIMUM	MINIMUM WALL THICKNESS
100	107.06	± 0.22	106.84	107.28	3.06
150	159.38	± 0.28	159.10	159.66	4.58
200	213.36	± 0.30	213.06	213.66	6.10
250	266.70	± 0.38	266.32	267.08	7.62
300	317.50	± 0.46	317.04	317.96	9.07
375	388.62	± 0.58	388.04	389.20	11.10
450	475.00	± 0.70	474.30	475.70	13.57
525	560.00	± 0.85	559.15	560.85	16.00
600	630.00	± 0.95	629.05	630.95	18.00
680	710.00	± 1.05	708.95	711.05	20.29

318.06.03 Ribbed, PVC Drainage Pipe Dimensions - The average inside diameter and minimum wall thickness of Ribbed PVC drainage pipe shall be as specified in Table 318-B when measured in accordance with ASTM D 2122.

TABLE 318-B PIPE DIMENSIONS RIBBED

INSIDE DIAMETER (mm)					
NOMINAL DIAMETER	AVERAGE	TOLERANCE	MINIMUM	MAXIMUM	MINIMUM WATERWAY WALL THICKNESS
450	448.31	± 1.41	446.90	449.72	3.30
525	527.05	± 1.50	525.55	528.55	4.06
600	596.90	± 1.78	595.12	598.68	4.58
680	673.10	± 2.00	671.10	675.10	5.20
760	749.30	± 2.30	747.00	751.60	5.96
900	901.70	± 2.70	899.00	904.40	7.36
1000	1003.30	± 3.00	1000.30	1006.30	8.26
1200	1206.50	± 3.55	1202.95	1210.95	10.16

318.06.04 Perforations - Perforated pipe up to the nominal diameter of 37.5 mm shall be type PSM and perforated in accordance with Figure I and Table 318-C. The perforations shall be 8 to 10 mm in diameter, circular, and cleanly cut. The gasketted bell and spigot end of the pipe shall be unperforated for a length equal to the depth of the socket and/or shoulder.

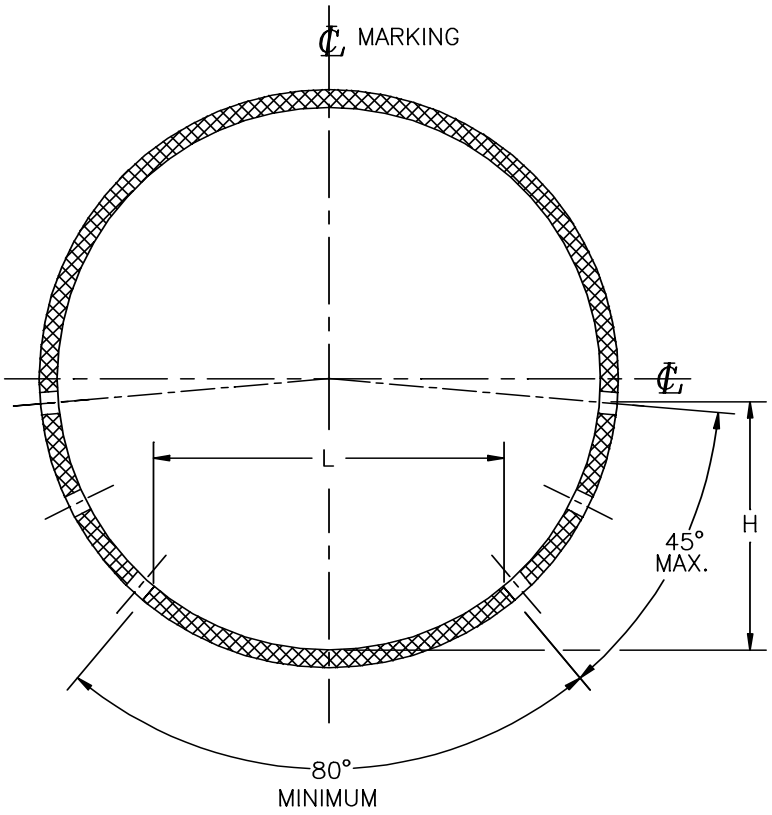


FIGURE I

TABLE 318-C PERFORATION REQUIREMENTS

NOMINAL DIAMETER (mm)	MINIMUM ROWS OF PERFORATIONS	HORIZONTAL HOLE SPACING (mm)	H (mm) *	L (mm) **
100	2	80 ± 5	30	65
150	4	80 ± 5	70	73
200	4	80 ± 5	93	130
250	4	100	115	163
300	6	150	140	195
375	6	150	175	240

*H: maximum height of the centreline of the uppermost perforations above the Invert.

**L: minimum length of the unperforated inner surface at the bottom of the pipe.

318.07 Physical Properties

318.07.01 Impact Strength - The minimum drop weight impact strength for pipe shall meet the requirements of Table 318-D when tested in accordance with DB SS 318.09.01.

318.07.02 Pipe Stiffness - The minimum pipe stiffness when tested in accordance with DB SS 318.09.02 and ASTM D 2412 shall be 320 kPa.

318.07.03 Pipe Flattening - There shall be no evidence of splitting, cracking or breaking when the pipe is tested in accordance with DB SS 318.09.03.

318.07.04 Extrusion Quality - The pipe shall not flake or disintegrate when tested in accordance with ASTM D 2152.

TABLE 318-D IMPACT STRENGTH AT 0ºC

NOMINAL DIAMETER (mm)	joules	ft. / lb.
100	93	69
150	147	108
200	175	129
250	190	140
300	205	151
375	230	170
450 and larger	250	184

318.08 Marking - All pipe shall be clearly marked at intervals of no more than 1.5 m with 5 mm or larger letters with the following information:

- Manufacturer's name or trademark
- Nominal diameter
- Material designation and cell class: PVC-12454-B

- The word "Drainage" or similar
- The class: SDR35 or PS320
- The applicable specification designation (ASTM D 3034 or F 794)
- Date of manufacture and plant designation

The marking on perforated pipe shall be 180º from a point equidistant between the bottom row of holes as shown in Figure I and Table 318-C.

318.09 Test Methods

318.09.01 Impact Resistance - Five specimens, each nominally 150 mm long for the PSM pipe and 300 mm long for the ribbed pipe, shall be tested in accordance with ASTM D 2444, using a 15 kg Tup "A" and the flat plate holder "B". The height of the drop shall not be less than 600 m or more than 3 m. The test shall be performed at 0°C (± 0.5°C), allowing no more than 15 seconds to complete the test after removing the sample from the conditioning medium. Perforated samples shall be positioned as shown in Figure I. Any cracking or other signs of material fatigue evident after the test shall constitute a failure. All five specimens must pass. If one specimen fails, five more specimens shall be tested. Nine out of ten specimens passing will be acceptable.

318.09.02 Pipe Stiffness - The pipe stiffness shall be determined at 5% deflection of the initial inside diameter in accordance with ASTM D 2412. Three specimens, each with a length equal to one pipe diameter shall be tested; all shall meet the required stiffness. Perforated samples shall be positioned as shown in Figure I.

318.09.03 Pipe Flattening - Three specimens, each nominally 150 mm long, shall be flattened between parallel plates in a suitable press until the distance between the plates is 40% of the original nominal diameter of the pipe. The loading rate shall be uniform and such that the compression is completed within two to five minutes. Remove the load and examine the specimen. All three specimens shall meet the requirements.

318.09.04 Pipe Dimensions - Measurements shall be taken in accordance with ASTM D 2122.

318.10 Inspection

a) The Quality Manager and Ministry shall have access to the fabrication plant for inspection and every facility shall be extended for this purpose.

b) Inspection may include checks on physical dimensions and the quality of work.

c) For the purpose of sampling for DB SS 318.07 and DB SS 318.09, one pipe from each lot of 50 pipes or fraction thereof in a shipment shall be selected for sampling. All test specimens shall be cut from the selected pipe(s).

d) If a test sample fails to meet the requirements of DB SS 318.07 when tested according to DBSS 318.09, that batch will be rejected and shall be replaced by the Supplier with a new batch that conforms to this specification at no additional cost.

318.11 Quality Assurance

a) The Supplier shall develop and maintain an effective quality control system to ensure that adequate inspection coverage is maintained throughout the manufacturing process. Evidence of such inspection shall be available to the Quality Manager and the Ministry prior to shipment from the place of manufacture or from the Supplier's storage facility.

b) The Contractor shall, and the Ministry reserves the right to, audit the Supplier's quality control system to ensure conformance with this specification.

SECTION 320

CORRUGATED STEEL PIPE

320.01 Scope - The Section covers the material and fabrication requirements for corrugated steel pipe and structural plate corrugated steel pipe.

Abbreviations:

CSP - Corrugated Steel Pipe: Galvanized sheet steel formed to finished shape by the fabricator.

SPCSP - Structural Plate Corrugated Steel Pipe: Hot rolled sheets or plates, corrugated, hot-dip galvanized, curved to radius, assembled and bolted together to form large pipes, pipe-arches, and other shapes.

320.02 Materials

320.02.01 Galvanized Steel Sheet - All culverts supplied under this specification shall be fabricated from steel sheets meeting all requirements of ASTM A 444 "Standard Specification for Zinc Coated (Galvanized) Iron or Steel Sheets for Culverts and Underdrains," with the following exceptions and additions:

320.02.01.01 - Steel shall conform to the chemical composition as specified in Table 320-A

TABLE 320-A CHEMICAL COMPOSITION

ELEMENT	HEAT ANALYSIS (MAXIMUM) %
Carbon	0.15 (except SPCSP)
Manganese	0.60 (except SPCSP)
Phosphorous	0.04
Sulphur	0.05

320.02.01.02 - Each sheet, or each 600 to 1500 mm of coil or strip shall be identified by showing the following:

i) Brand or name of sheet manufacturer (mill).

ii) "CSP Sheet/Tole TTAO"

iii) Nominal overall zinc coated sheet thickness in mm.

iv) Heat and Coil numbers.

v) Zinc coating designation.

vi) "ASTM A 444" or "CSPI-501-M" of "CSA G401-M."

Exceptions and additions to ASTM A 444 requirements:-

320.02.01.03 The requirement for the sheet manufacturer to run tensile tests on samples from each cast or heat (See ASTM A 444 Paragraph 5.2) will be waived if the sheet or plate is authorized to be stamped "CSPI-501-M."

320.02.01.04. - In addition to the Coating Adherence requirements of ASTM A 444 Paragraph 6, the Finish and Adherence of casting requirements of ASTM A 123, Paragraph 9 shall also apply.

320.02.01.05. - The mass of zinc coatings, total both sides shall be not less than Z610 (610 g/m²) when tested by the Triple Spot Test or not less than 550 g/m² by the Single Spot Test.

320.02.01.06. - Thickness and Tolerances shall conform to the requirements of Table 320-B.

TABLE 320-B CSP SHEET THICKNESS AND TOLERANCES

CSP SHEET - THICKNESS AND TOLERANCES (mm)		
Nominal Overall Thickness[1] (inc. Zinc)	Tolerance on Nominal Overall Thickness	Design Base Metal Thickness[2]
1.0	±0.13	0.82
1.3	±0.15	1.12
1.6	±0.18	1.40
2.0	±0.18	1.82
2.8	±0.20	2.64
3.5	±0.23	3.35
4.2	±0.23	4.08

[1] Thickness shall be measured at a point on the sheet not less than 10 mm from an edge. Field